

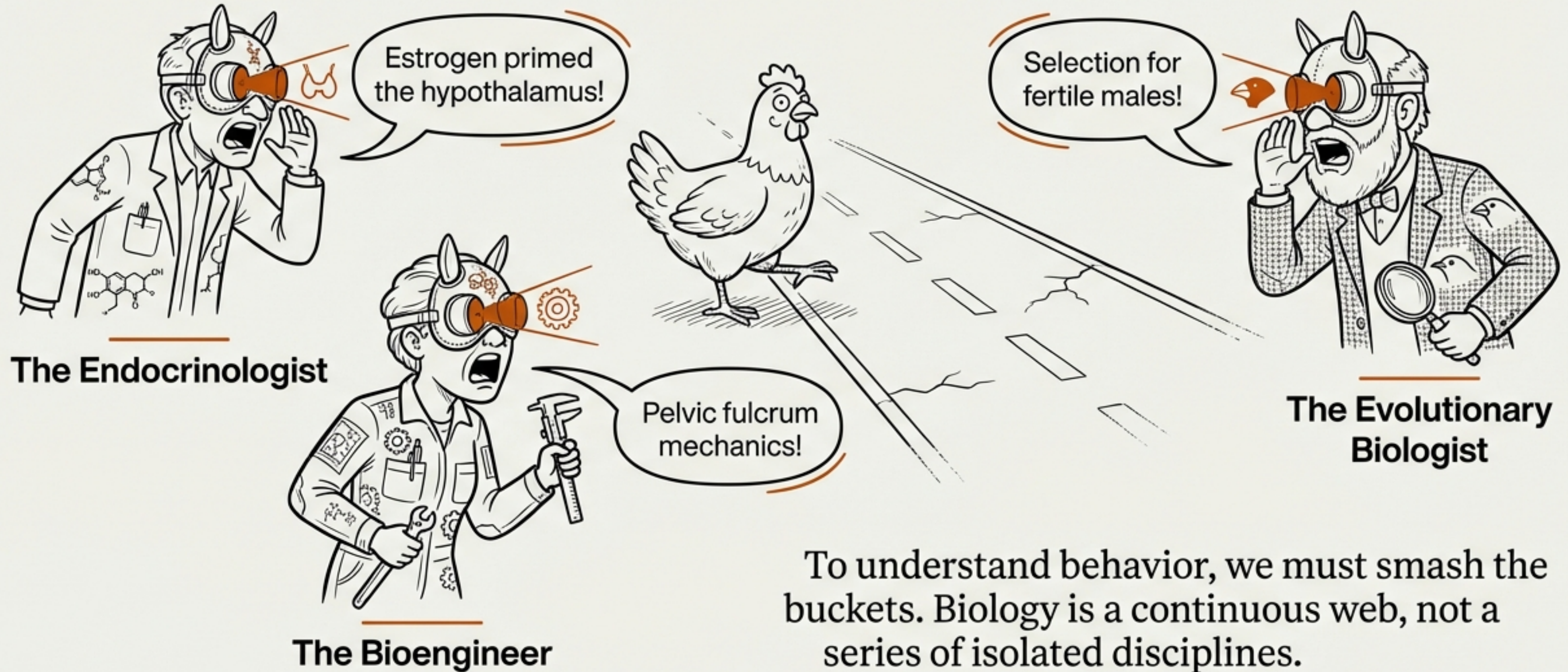


The Biology of Humans at Our Best and Worst.

A forensic investigation into the seconds, minutes, and days preceding a single human action.

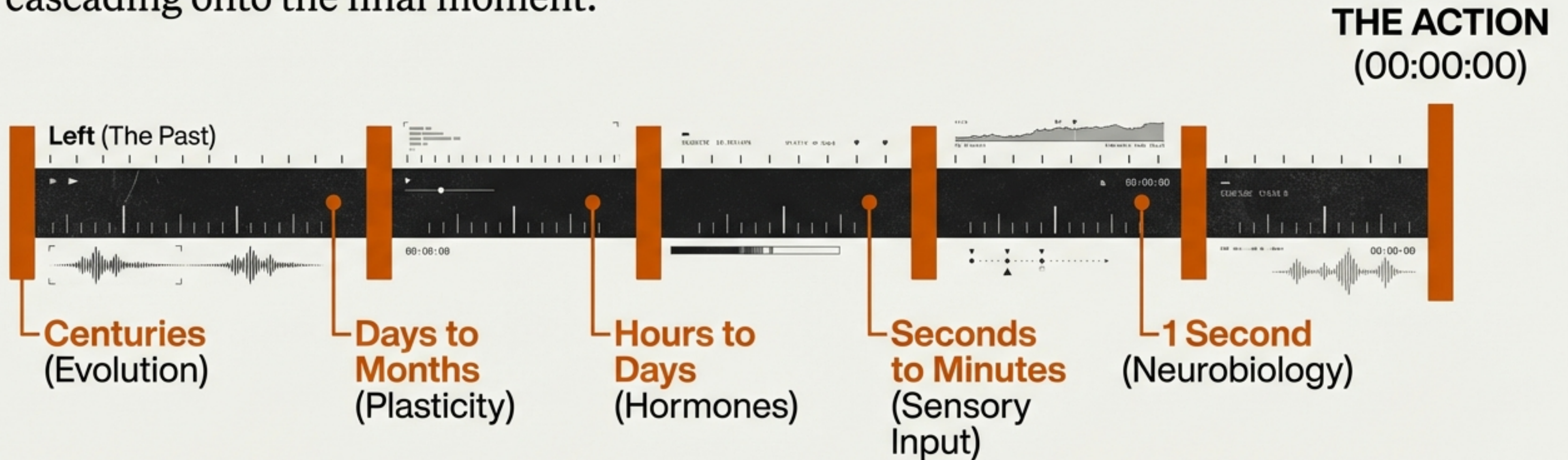


The Trap of Categorical Thinking



A Forensic Timeline of Behavior.

A behavior is not caused by a gene, a hormone, or a childhood trauma. It is the result of all of them cascading onto the final moment.





The Anatomy of an Action.

The Biology:

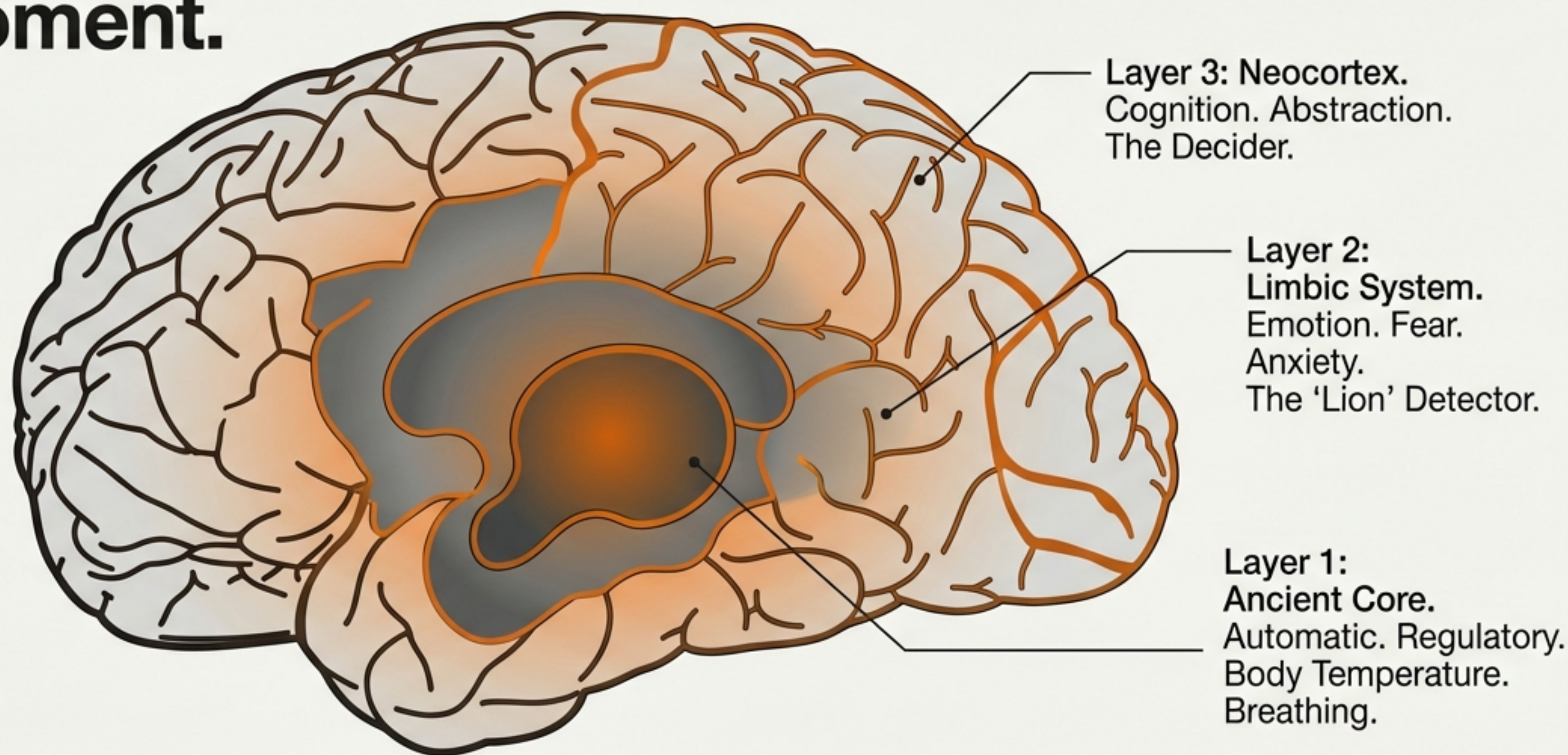
The motor cortex commands the muscle to contract. The physiology is identical.

The Meaning:

Defined entirely by context. Aggression and Empathy often use the same machinery.

[TIME: -1 SECOND]

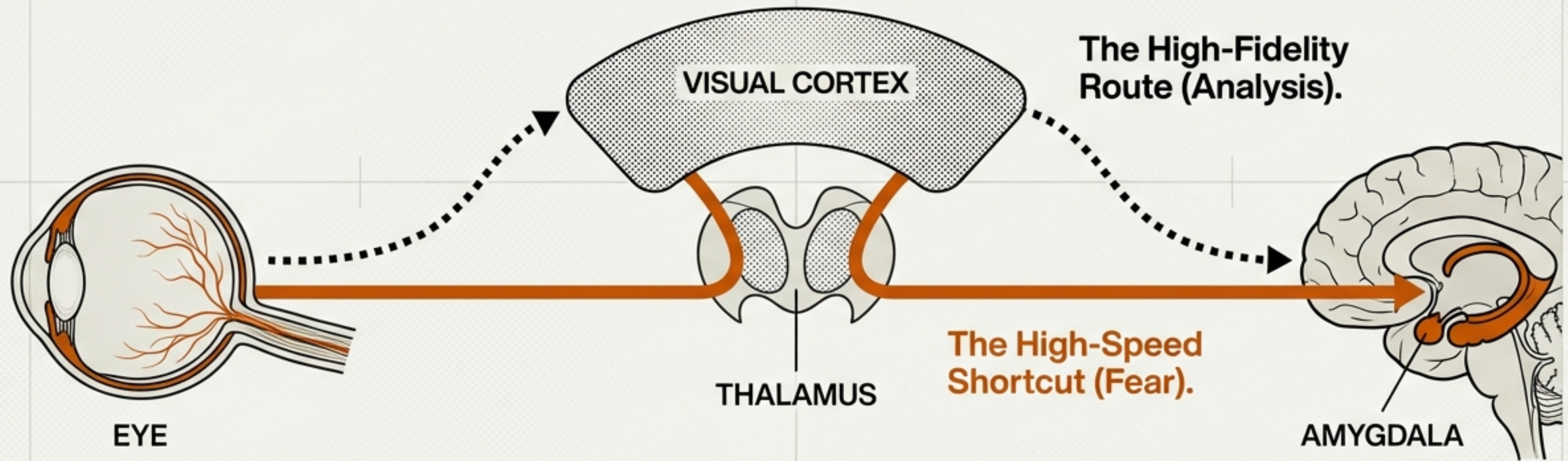
The Machinery of the Moment.



A useful metaphor for the internal civil war between impulse and decision.

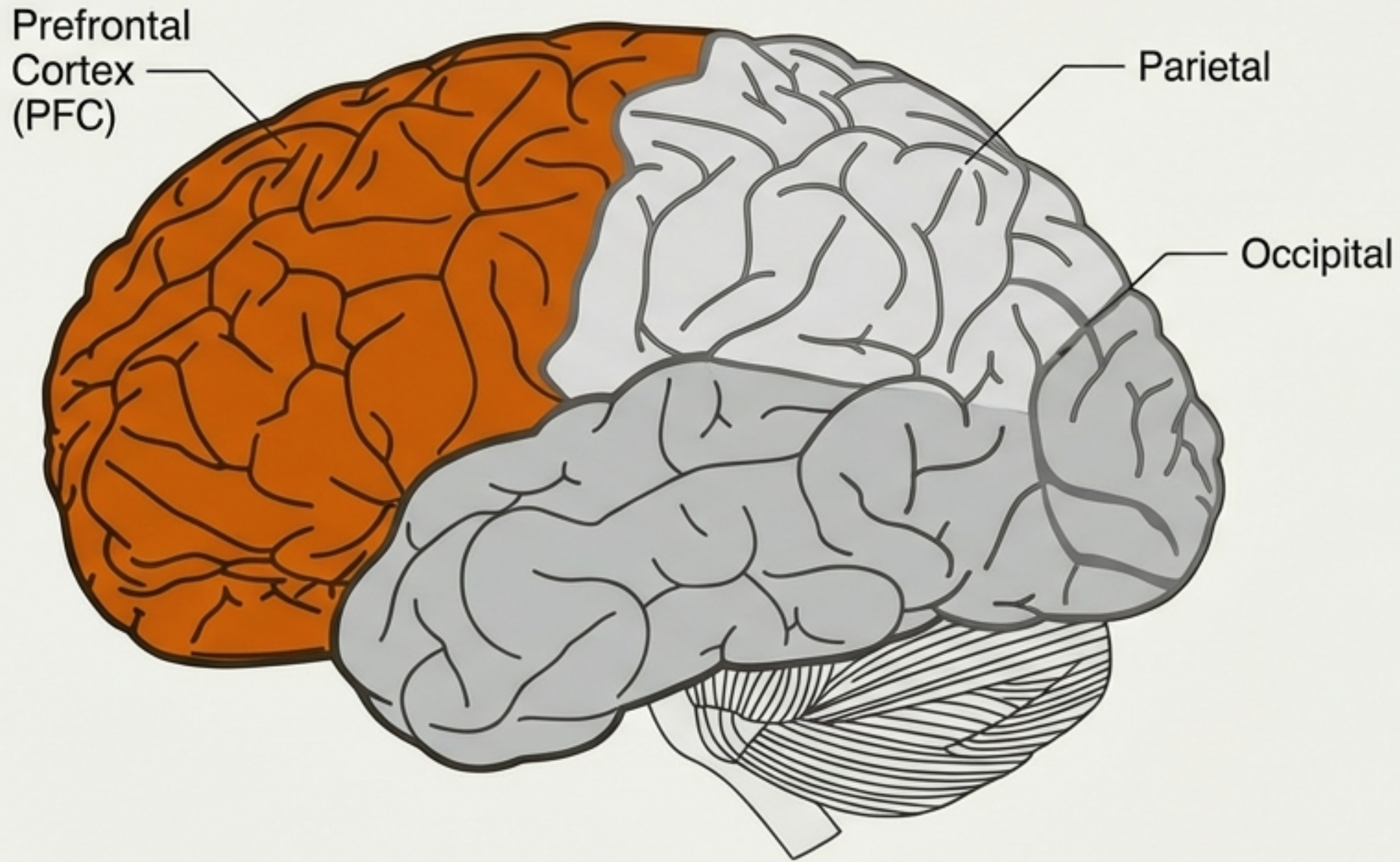
[TIME: -1 SECOND]

The Amygdala: The Alarm Bell.



The Amygdala asks: "Is it a snake?" The Cortex says (seconds later):
"No, it's a stick." But your heart is already racing.

The Frontal Cortex: The Decider.

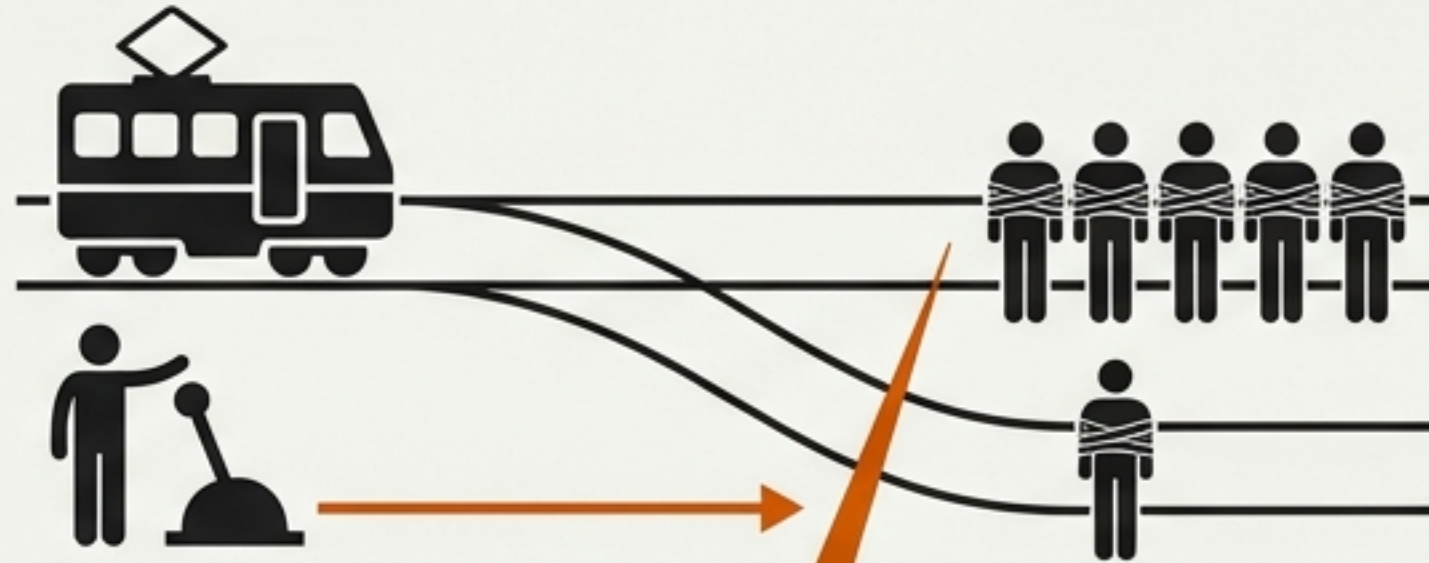


Phineas Gage (1848). A tamping iron destroyed his Frontal Cortex. He survived the wound but lost his emotional regulation. He was "no longer Gage."

The PFC is the "brake" on the Amygdala's "accelerator." It makes you do the harder thing when it is the right thing to do.

Logic vs. Viscera: The Trolley Problem

Scenario A



Active Region:
Dorsolateral PFC (Logic)

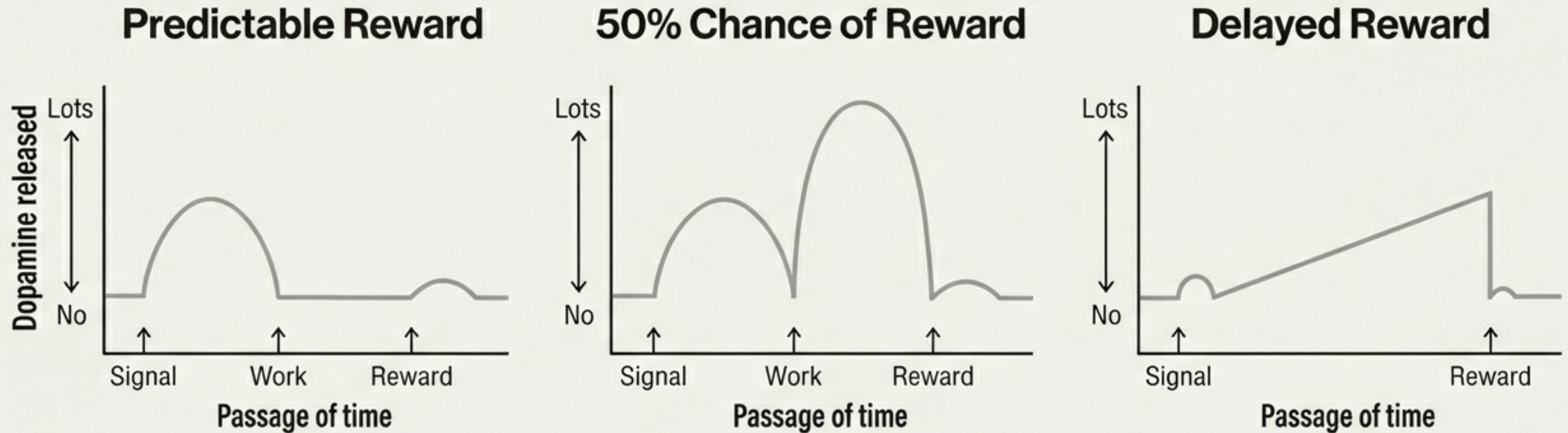
Scenario B



Active Region:
Amygdala & Ventromedial PFC (Emotion)

The math is the same.
The biology is different.

Dopamine: The Engine of Anticipation.



Dopamine is not about pleasure. It is about the pursuit of pleasure.
Uncertainty fuels the addiction.

[TIME: -10 MINUTES]

The Sensory World.

The brain does not act in a vacuum. **Before** we are **conscious** of a **threat**, the **Fusiform Face Area** processes race and gender in milliseconds.

P200 Waveform = Threat Signal

Often higher for “Out-Group” faces.



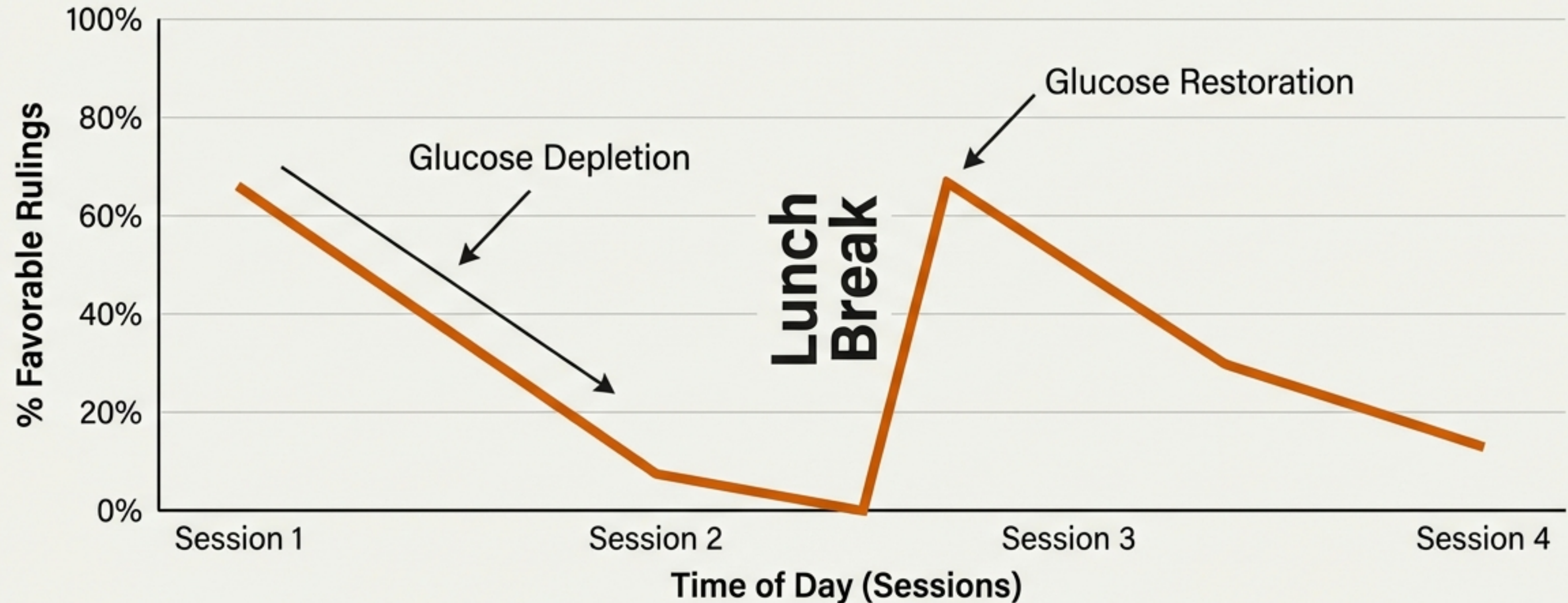
The Visual Bias: Gun or Cellphone?



The “Shooter Bias” (Correll, University of Colorado): Subjects are faster to shoot an armed black target than an armed white target, and more likely to mistakenly shoot an unarmed black target.

The brain notes *the color of skin, real fast.*

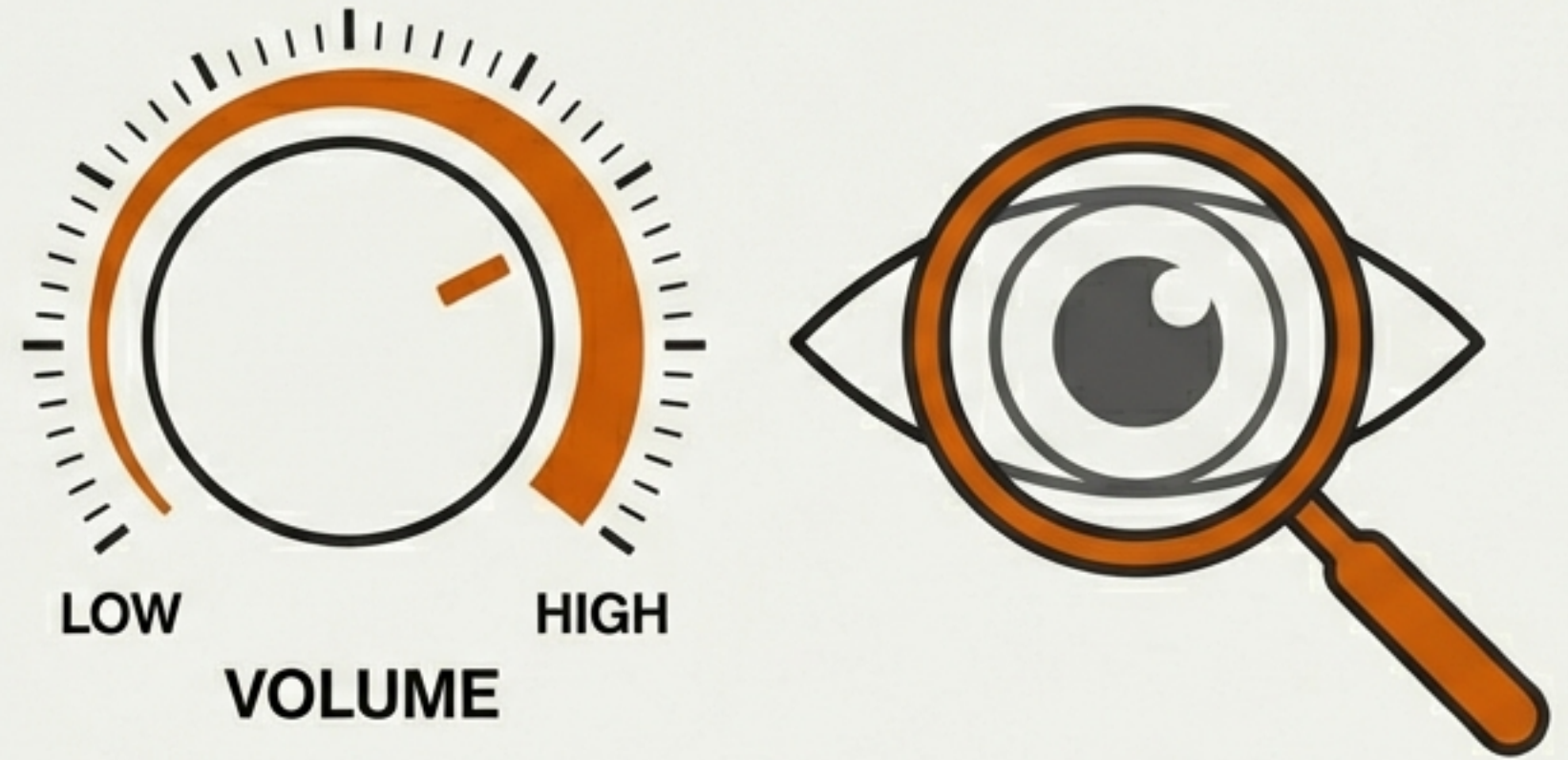
Interoception: The Hungry Judge.



Justice is not blind; it is metabolic. When energy is low, the brain defaults to the "easier" answer: Deny parole.

[TIME: -24 HOURS]

Hormones: The Volume Knob.

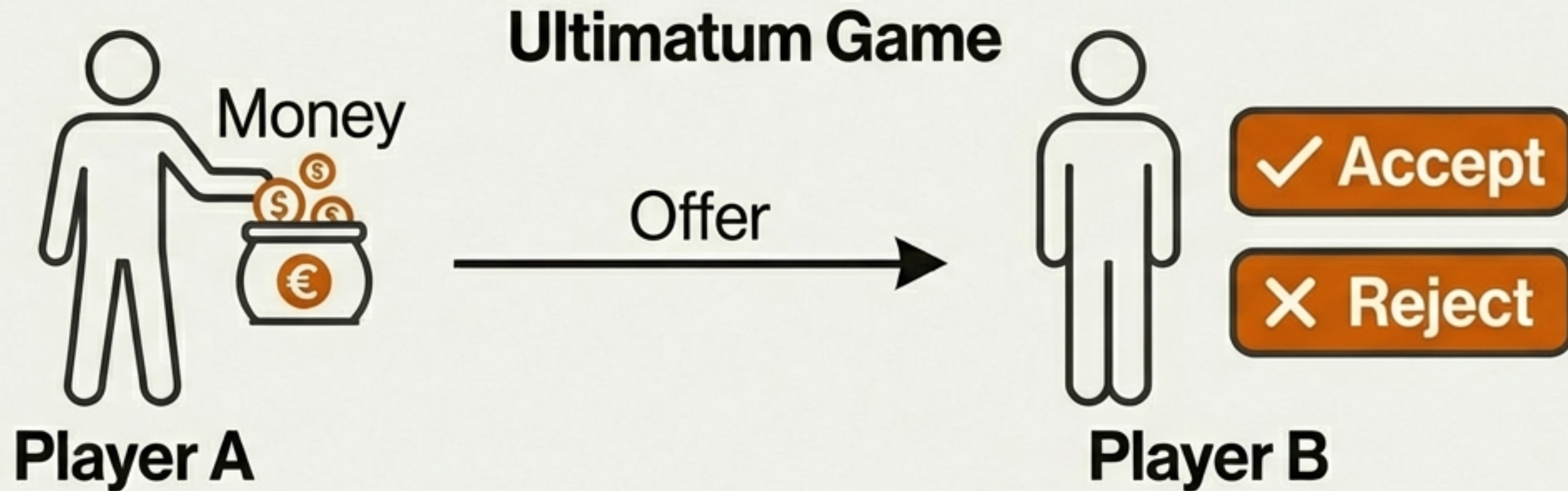


Hormones do not cause behavior. They alter sensitivity.

- If you are listening to aggressive music, turning up the knob makes it louder.
- If you are listening to peaceful music, turning up the knob makes that louder too.

Myth: Testosterone Causes Aggression.

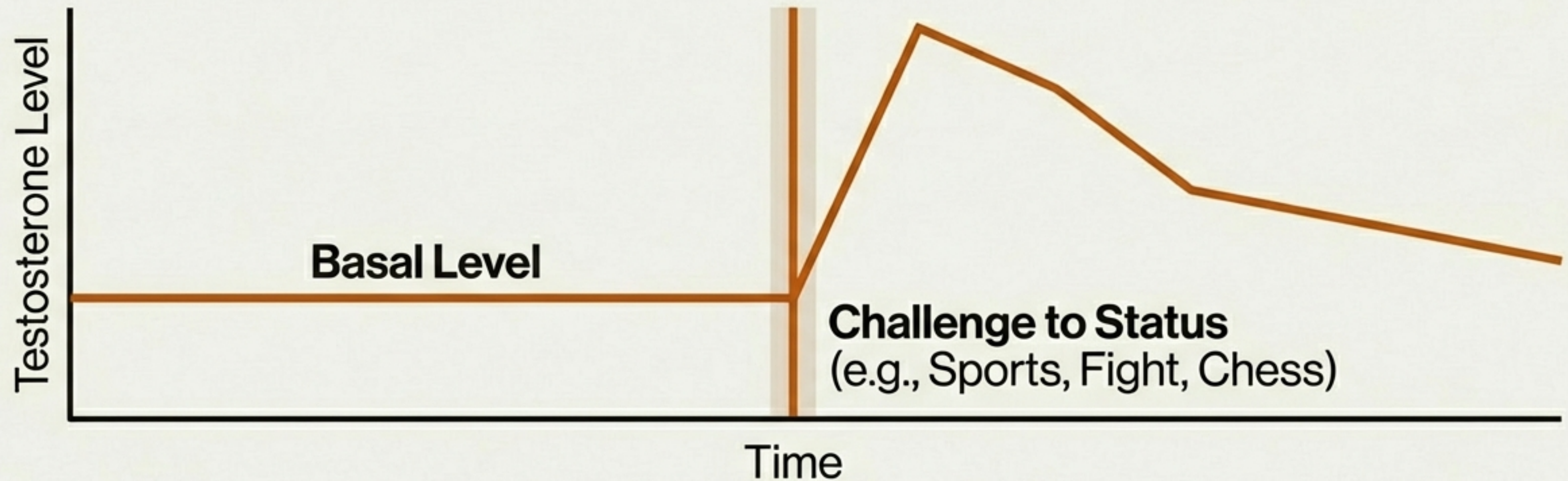
Reality: Testosterone Causes Status-Seeking.



The Eisenegger & Fehr Study: When status is maintained by being fair, testosterone makes people *more* generous.

Testosterone amplifies whatever behavior is rewarded by the social structure.

The Challenge Hypothesis.



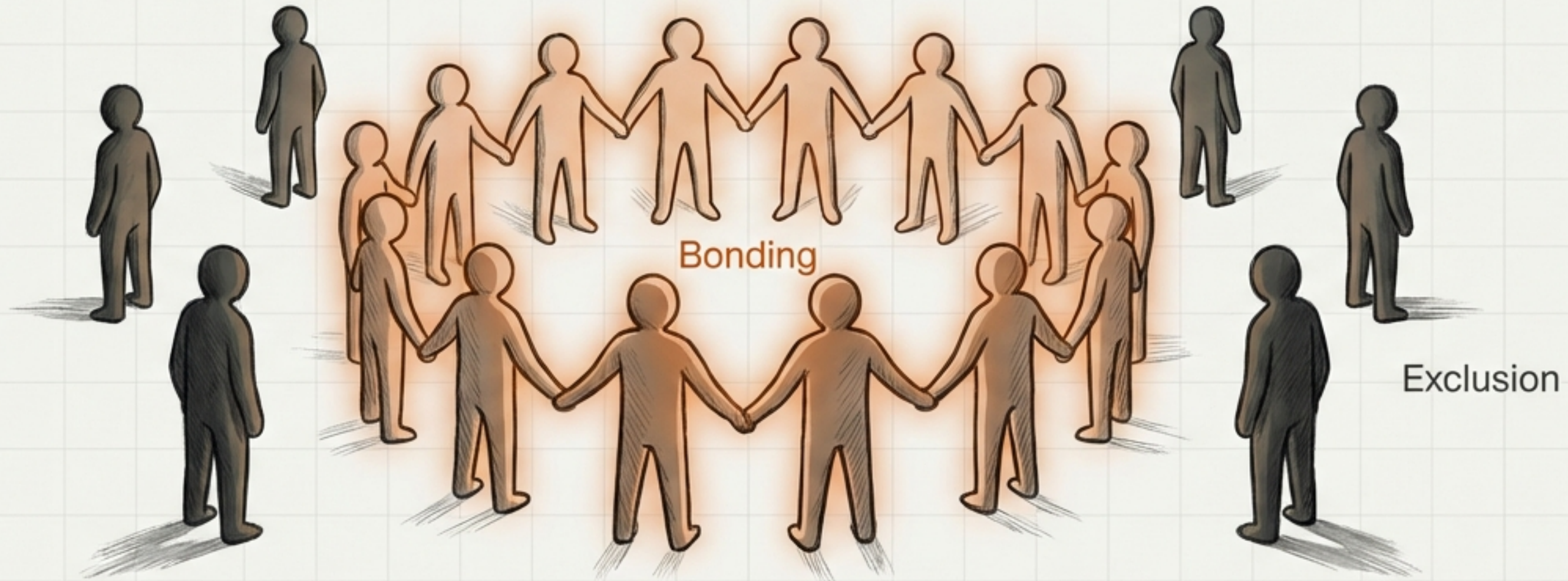
Testosterone levels rise specifically during a challenge. This rise amplifies preexisting tendencies.

The Talapoin Monkey: Testosterone didn't make the middle-ranking male attack the boss. It made him a bigger bully to the guys below him.

Oxytocin: The ‘Us vs. Them’ Molecule

Myth: The “Love Molecule”.

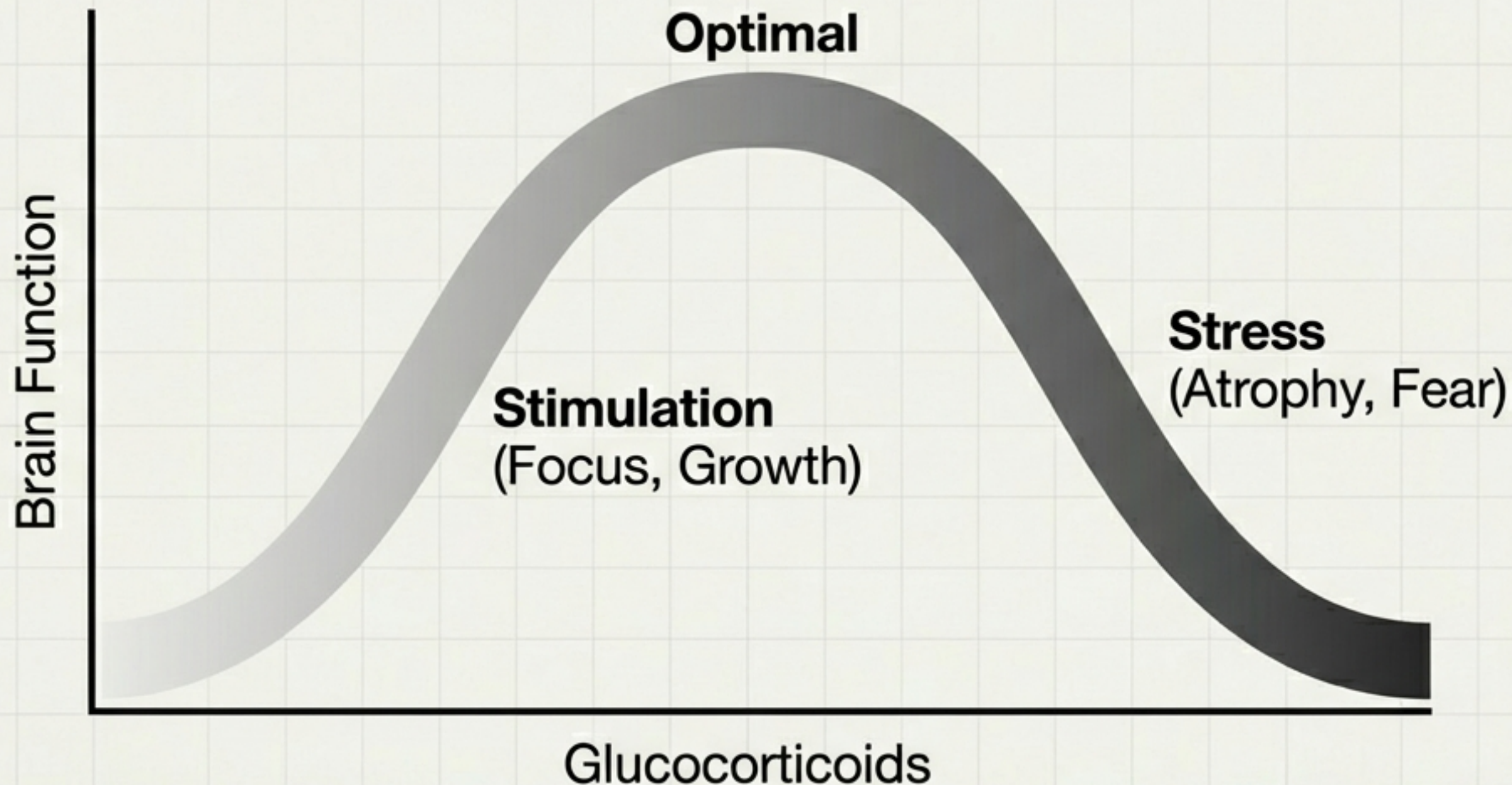
Reality: Oxytocin fosters parochial altruism. It makes us kinder to “Us” and more aggressive to “Them”.



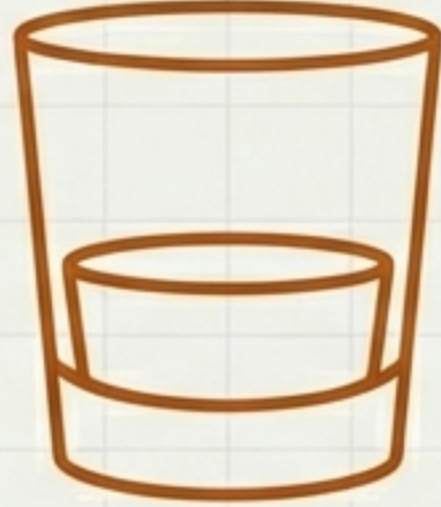
De Dreu Study: Subjects on Oxytocin were more likely to sacrifice a person with an “Out-Group” name.

The Biology of Stress.

Chronic stress atrophies the Prefrontal Cortex and hypertrophies the Amygdala. The brain becomes wired to perceive threats where none exist.



The Alcohol Interaction.

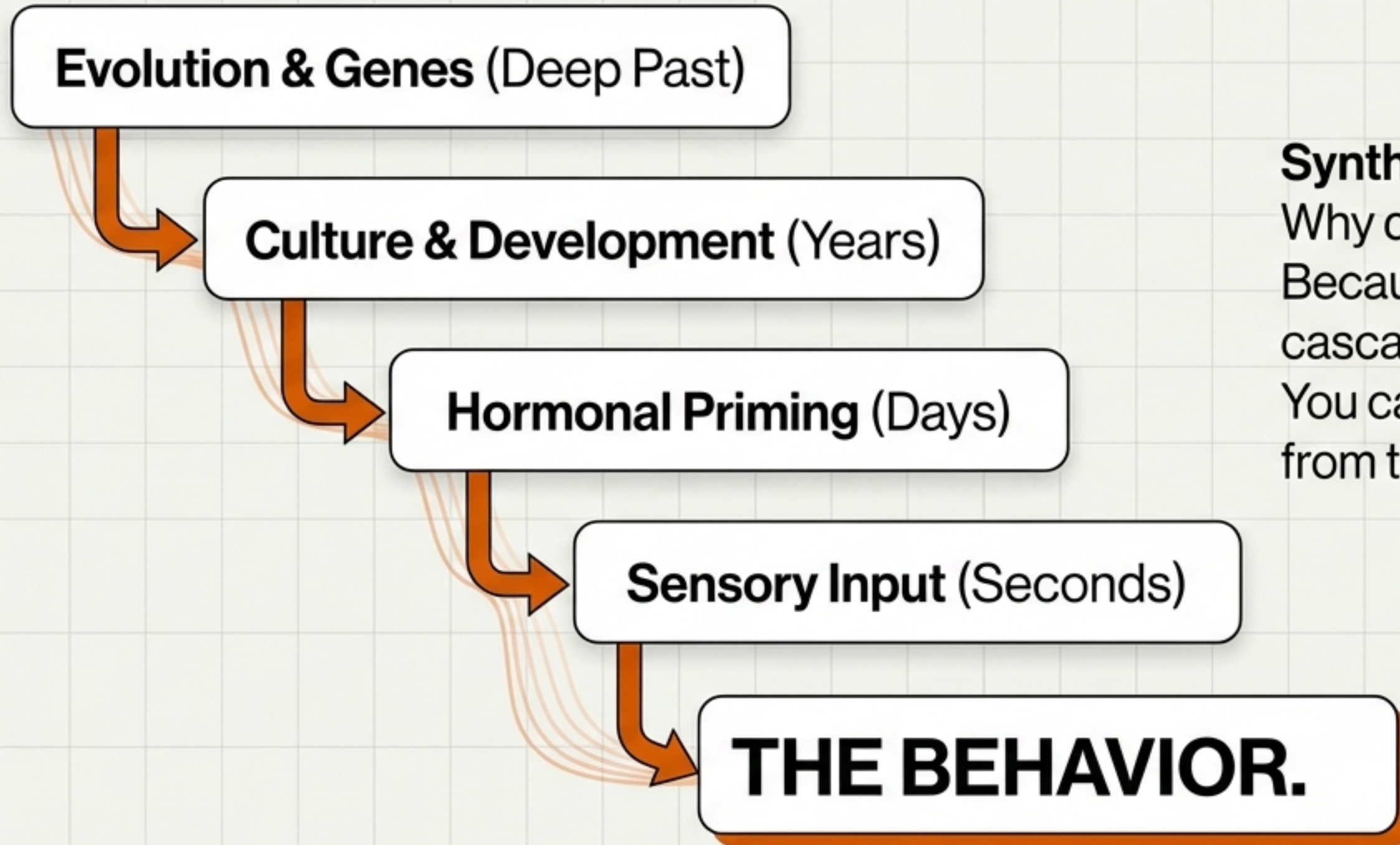


Alcohol is a solvent for social learning.

Alcohol does not inherently cause aggression. It triggers aggression in those who *believe* alcohol makes them aggressive.

In cultures where drunkenness implies sleepiness, violence does not increase.

No Buckets. Only History.



Synthesis text

Why did the behavior happen?
Because of all of these factors,
cascading onto the final moment.
You cannot separate the biology
from the environment.

The Opposite of Love is Indifference.



“Understanding the biology of violence doesn’t mean forgiving it. But it separates ‘evil’ from biological malfunction. Context is everything.”

Based on ‘Behave’ by Robert M. Sapolsky.